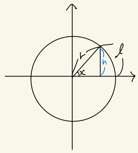
<인공지능 2주차 1차시 수업>

Math
- Tech
- 개념
- 기하학

$e=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \to \frac{1}{n}=t,\ \lim_{t\to 0}(1+t)^{\frac{1}{t}}$



$l = rx$

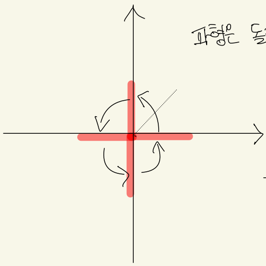
$S=\frac{1}{2}rl=\frac{1}{2}r^2\cdot x$

$\sin x = \frac{h}{r}$

$\lim_{x\to 0}\frac{\sin x}{x} = \lim_{x\to 0}\frac{h}{r\cdot x} = \lim_{x\to 0}\frac{h}{l} = 1$

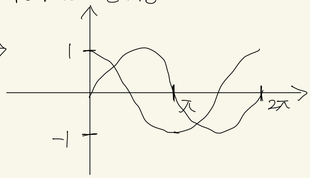
$\lim_{x\to 0}\frac{x}{\sin x} = \lim_{x\to 0}\frac{1}{\frac{\sin x}{x}} = 1$

$\lim_{x\to 0}\frac{\tan x}{x}=1 \quad \lim_{x\to 0}\frac{x}{\tan x}=1$



파형은 돌리지만

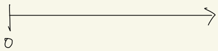
Real Term만 사용



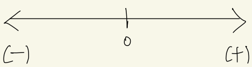
$\sin\left(\theta+\frac{\pi}{2}\right) = \cos\theta$

$\cos\left(\theta-\frac{\pi}{2}\right) = \sin\theta$

$\sin(\theta+\pi) = -\sin\theta$

① 

1D Scalar

② 

1D Vector

$\left.\begin{matrix}\text{Scalar}\\ \text{Vector}\end{matrix}\right\} \in$ Real Number

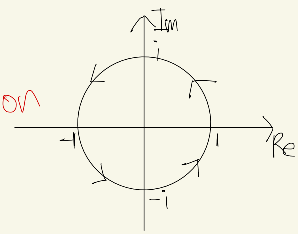
실수: Real Number

허수: Imaginary Number

③ $i^2=-1,\ i^3=i^2\cdot i=-i,\ i^4=1$

↳ 모른다

2D Rotation



$Z=a+bi$: 복소수(Complex)

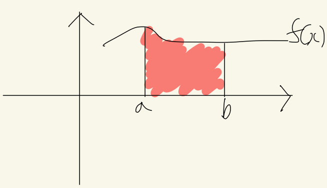
부정적분 $\int f(x)\,dx = F(x)+C$

ex) $\int \frac{1}{x}dx = \ln x + C,\ \int \cos x\,dx = \sin x$

$\int \sin x\,dx = -\cos x,\ \int \sec^2 x\,dx = \tan x$

$\int \sec(x^2)\tan(x^2)\,2x\,dx = \sec x^2 + C$

정적분 $\int_a^b f(x)\,dx = F(b)-F(a)$



ex)

$y=\ln(\tan(\cos x))$

$$\int \frac{-\sec^2((\cos x))\cdot \sin x}{\tan(\cos x)}\,dx = \ln|\tan(\cos x)|+C$$

$y=\ln(\sec(e^{\ln x}))$ $\quad y'=\displaystyle\int \frac{\sec(e^{\ln x})\cdot\tan(e^{\ln x})\cdot e^{\ln x}\cdot\frac{1}{x}}{\sec(e^{\ln x})}\,dx$

$$\int \frac{5x^4+6x+\frac{\cos x}{\sin x}}{x^5+3x^2+\ln(\sin x)}\,dx = \ln|x^5+3x^2+\ln(\sin x)|+C$$

$\int f\cdot g\,dx = f\cdot G - \int f'\cdot G\,dx$ / $\int x\sin dx = -\cos x + \int \cos x\,dx$

f · 미분 편한거

g · 적분 〃

$f=x,\ f'=1$

$g=\sin x \quad G=-\cos x$

$= -x\cos x+\sin x+C$

$-\cos x+(-x)(-\sin x)$
$+\cos x+0$
$= x\sin x$

100, 200, 300 $\quad \bar{x}=M_x=E[X_i]$

$Avg=\dfrac{\sum_{i=1}^{N} x_i}{N}$

	편차
100	$-100 \to \square^2$
$\boxed{200}$	0
300	100

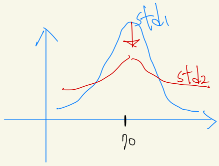
평균편차 : 0

음 → 양
① | |
② $\square^2$
③ e

$\sqrt{Var}$ = 표준편차 (Standard Deviation)
= Std

Variance (분산) : 편차의 제곱을 평균

$Var=\dfrac{\sum_{i=1}^{N}(x-\bar{x})^2}{N}$ 평균



Std1 < Std2 → 얼마만큼 퍼졌냐.

$y=f(t)$
$t=0$, 100마리
$t=1$, 300마리
$t=10$, ?

$\dfrac{df(t)}{dt}=k\cdot f(t)$ ↱ 비례
↓
1차 미분방정식

$\dfrac{df(t)}{dt}=kf(t) \Rightarrow \dfrac{df(t)}{f(t)}=k\cdot dt$

$\int \frac{1}{f}\,df=\int k\,dt$

← $\ln|f|+C_1=kt+C_2$

$\ln|f(t)|=kt+C$

$|f(t)|=e^{kt+C}=e^{kt}\cdot e^{C}$

$f(t)=\pm e^{C}\cdot e^{kt}=Ce^{kt}$

$f(t)=Ce^{kt}$

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x → scalar → Vector/Metric

"Chain Role"

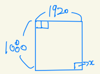
$$x \xrightarrow{\text{Tone}} \boxed{A} \xrightarrow{h=A(x)} \boxed{B} \xrightarrow{g=B(x)} \boxed{C} \xrightarrow{i=C(g)} \Rightarrow : \text{Forward Propagation}$$

$\frac{\partial i}{\partial x} = ?$ $i = C(g) = C\{B(h)\} = C\{B[A(x)]\}$

Loss ⟸ Backward Propagation.

$$\frac{\partial i}{\partial x} = \frac{\partial i}{\partial g} = \frac{\partial g}{\partial h} = \frac{\partial h}{\partial x}$$

full-HD



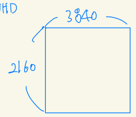
ex) $x^{10\times10}$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_{10} \end{bmatrix}$$

2D

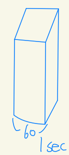
$$x = \begin{bmatrix} x_1 & \cdots & x_{10} \\ \vdots & & \vdots \\ x_{91} & \cdots & x_{100} \end{bmatrix}$$

4K = UHD



UHD · 60Hz 1초에 60장

[장수] [Low] [Col]



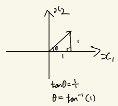
행 열
[ ] [ ]
[Low] [Col]
[y] [x]
y index x index

X 안좋음

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\|x\| = \sqrt{2}$

$\angle x = \tan^{-1}(1) = \frac{\pi}{4}$



$\tan\theta = 1$

$\theta = \tan^{-1}(1)$

① Scalar : $x \in R$

② Vector : $x \in R^m$ → m×1 → col ⟹ $x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$ → $x_i \in R$ $(i = 1 \sim m)$

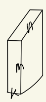
③ Metrix : $x \in R^{m\times n}$ ⟹ $x = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}$ → $x_{ij} = \begin{bmatrix} i = 1 \sim m \\ j = 1 \sim n \end{bmatrix}$

x_1 x_2 x_n

$= [x_1 \cdots x_n], \; x_k \in R^m$

$k = 1 \sim n$

④ Tensor : $x \in R^{m\times n\times k}$



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① Unit Vector $x \in R^2$ $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\|x\|_2 = 5 \Rightarrow U_x = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$

Norm

$x \in R^n$ $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

② Bass Vector

$W_1x_1 + W_2x_2$
= Weighted Sum
= Linear Combination (선형 조합)
↳ 과정을 span이라함 같은 말.

ㄱ ㄴ ㄷ ㄹ ··· (자음)
ㅏ ㅑ ㅓ ㅕ ··· (모음)

공 Vector space
↑
closed

ㄱ → ㄲ (가능) Linear Dependent
ㄱ → ㄴ (불가능) 그러나 새 Linear Independent 하다.
고유의 자음과 모음을 Bass Vector라고 부름

L1 norm $= l_1 = \Sigma|x_i| = \|x\|_1$

L2 norm $= l_2 = \sqrt{\Sigma x_i^2} = \|x\|_2$

↳ $\|x\|_2^2 = \Sigma x_i^2 = \|x\|^2$

x_1
x_2
→ $R^2 = \sqcup$ $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\boxed{W_1x_1 + W_2x_2} \Rightarrow$ Span

$$\text{Span}(x_1, x_2) = \{\underbrace{W_1x_1 + W_2x_2}_{\text{Linear C}} \mid W_1, W_2 \in R\}$$

$x_1 \sim x_n$

$W_1x_1 + W_2x_2 \cdots + W_nx_n$

$W_1 \cdots W_n = 0$ 선형독립

$W_1x_1 + W_2x_2 = 0$

$W_1x_1 = -\underbrace{W_2}_{W_2}x_2$

$W_1x_1 = W_2x_2$

$x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$
↓
Rank(x) = 2 (Bassis Vector의 개수)

$x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in R^3$
⇓
Rank(x) = 3
full Rank

full Rank가 아니면 inverse Metric 존재X

ex) if (k != full Rank)
k^{-1} = None.

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Col $\cdot\ x \in R^m$

Metric $\cdot\ x \in R^{m \times n} \Rightarrow x = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & & \vdots \\ x_{1m} & \cdots & x_{mn} \end{bmatrix} = [x_1, x_2 \cdots x_n]$, $x_i \in R^m$

Tensor $\cdot\ x \in R^{m \times n \times k}$

ex) $x = \begin{bmatrix} 3 \\ -5 \\ 2 \end{bmatrix}$

$l_1 = 10$

$l_2 = \sqrt{3^2 + (-5)^2 + 2^2}$

Supervised Learning

A (정답): Label (ground truth)

B (예측)

$\sum_{i=1}^{N} \left| (A_i - B_i) \right| = L1$ loss Function (Loss)

$Loss = \sqrt{\sum_{i=1}^{N} (A_i - B_i)} = L_2$ "

parameter θ → 최솟값을 만들어주는 값

arg min Loss Function

$= \arg\min_{\theta} \left\{ \sum_{i=1}^{N} (A_i - B_i) \right.$

$\cdot$ Unit Vector

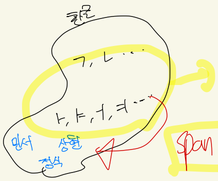
→ 크기 → Norm

$x \in R^m$

$x = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$

$l_1 = \sum |x_i| = \|x\|_1$

$l_2 = \sqrt{\sum x_i^2} = \|x\|_2$



Bassis

Span = Linear Combination (선형 조합)

→ Vector space O (Bassis 일 때)

→ Vector space X

$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$: x_1, x_2 Linear Independent

$\therefore\ span\{x_1, x_2\} = \{w_1 x_1 + w_2 x_2 \mid w_1, w_2 \in R\} \in R^2 = ✓$

⇒ Bassis Vector를 span했을 때 Vector space는 close되어있다.

$\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \frac{3}{2}\begin{bmatrix} 2 \\ 0 \end{bmatrix} + 0\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} = w_1\begin{bmatrix} 2 \\ 0 \end{bmatrix} + w_2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

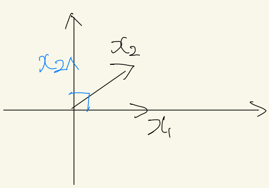
$= \begin{cases} 2w_1 + w_2 \\ w_2 \end{cases}$

$2w_1 + 4 = 3 \quad w_1 = -\frac{1}{2} \quad w_2 = 4$

$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$Span(x_1, x_2) \in R^2 = ✓$

직교 orthogonal



$w_1 x_1 + w_2 x_2 = w_3 x_3$

L. indep $\xrightarrow{(O)\ (X)}$ orthogonal

$\xleftarrow{O}$

$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}$ $x_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$Span(x_1, x_2, x_3) \in R^3$

Shall <Bassis Vector는 항상 Linear Idp>

= have to = must

$R^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $R^3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$x_1 \quad x_2 \quad x_3$

$x_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ → orthonormal (벡터의 크기가 1)

∴ Bassis Vector를 span했을 때 Vector space를 만들어냈다.

→ Bassis vector는 서로 Linear Independent 해야한다.

$A \in \mathbb{R}^{m \times n}$

$A \cdot B \Rightarrow C \in \mathbb{R}^{m \times k}$

$(m \times n)(n \times k)$

① Transpose

$A \in \mathbb{R}^{m \times n}$, $A^T \in \mathbb{R}^{n \times m}$

$m>n$ $m<n$ $m=n$

"rectangular" "square"

$(AB)^T = B^T \cdot A^T$

$(A^T)^T = A$

√ $(A+B)^T = A^T + B^T$

② Symmetry (대칭행렬)

$A = A^T$

$A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

$A = XX^T$

$A = A^T$

$(XX^T) = (\underbrace{X}_{A}\underbrace{X^T}_{B})^T = (X^T)^T \cdot X^T = XX^T$

③ Inverse (역행렬)

$x = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

full $Rank(x) = 2$ / Linear Independent

$x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 3 \\ 1 & 0 & 0 \end{bmatrix} \Rightarrow Rank(x) = 2$

full Rank를 가져야만 Inverse Metric이 존재.

=> 각 Bassis vector들이 서로 Linear Independent 해야함.

$AB = I \rightarrow$ 항등행렬 (Identity) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$B = A^{-1}$ $(AB)^{-1} = B^{-1} \cdot A^{-1}$

$(A^{-1})^{-1} = A$

√ $(A+B)^{-1} \neq A^{-1} + B^{-1}$

대각행렬

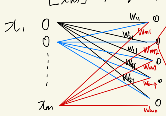
$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Zero Metric

$m=10$, $n=5$

$x \in \mathbb{R}^m = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}$, $y \in \mathbb{R}^n = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ $z \in \mathbb{R}^k$



$y_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + w_{41}x_4 + \cdots + w_{m}x_n$

y_2

y_3

y_4

y_5

$x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ · $W \in \mathbb{R}^{m \times n}$

$= y = W^T x \in \mathbb{R}^{n \times 1}$

$\Downarrow$

$(n \times m)(m \times 1)$

$z = W^T \cdot y$

$= W^T \cdot W^T x$

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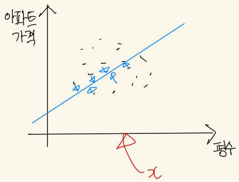


x_1, x_2

Linear Regression (선형회귀)

$A x = b \longrightarrow x = \cancel{A^{-1}} b$ square

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$\begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b \end{bmatrix}$

$Ax = b$

$A^T \cdot A x = A^T \cdot b$

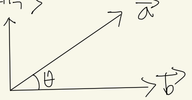
square

$(n \times m)(m \times n)$

$\therefore x = (A^T A)^{-1} \cdot A^T \cdot b$

<내적>

Inner Product (Dot)] Vector의 곱



$\in R^2$

$R^{m=10}$ $\vec{x}$ $x = \begin{bmatrix} x_1 \\ \vdots \\ x_{10} \end{bmatrix}$

$x = R^m, y = R^m$

$\boxed{\langle x, y \rangle = x^T \cdot y \quad (m\times1)(m\times1) \to (1\times m)(m\times1)} = \sum_{i=1}^{m} x_i y_i$

$(x_1 \cdots x_m) \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$

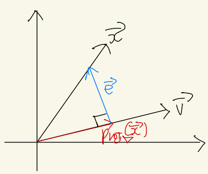
$\vec{a} \cdot \vec{b} = \langle \vec{a}, \vec{b} \rangle = \|\vec{a}\|_2 \|\vec{b}\|_2 \cos\theta$

$\theta = \frac{\pi}{2} =$ orthogonal

$\langle x, x \rangle = x^T \cdot x = \sum_{i=1}^{m} x_i^2 = \|x\|_2^2$

$\boxed{\therefore \|x\|_2^2 = x^T x}$

2 projection



i) $Proj_{\vec{v}}(\vec{x}) = \alpha \cdot \vec{v} \quad (\alpha \in R)$

ii) $\vec{Proj} + \vec{e} = \vec{x} \quad \therefore \vec{e} = \vec{x} - \vec{Proj}$

iii) $\vec{e} \perp \vec{v} \quad \therefore \langle v, e \rangle = \langle e, v \rangle = 0$

$= v^T e$

$= v^T (x - Proj)$

$= v^T x - v^T Proj$

$= v^T x - \alpha v^T v = 0 \quad \therefore \alpha = \frac{v^T x}{v^T v}$

$Proj_{\vec{v}}(\vec{x}) = \alpha \vec{v}$

$= \frac{v^T x}{v^T v} \cdot v$

$= \frac{v v^T}{v^T v} x$

$= \frac{\vec{v} \cdot \vec{v}^T}{\|v\|_2^2} \vec{x}$] Inner Product

3 벡터 편미분

① $y = Ax, \; y \in R^m, \; A \in R^{m \times n}, \; x \in R^n$

$\boxed{\Rightarrow \frac{\partial y}{\partial x} = \frac{\partial Ax}{\partial x} = A}$

$\begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & & a_{2n} \\ \vdots & & \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

$y_1 = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n$

$y_2 = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n$

$\vdots$

$y_m = a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n$

$y_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n$

$(i = 1 \sim m)$ $\quad \frac{\partial y}{\partial x}$

$\Rightarrow y_1 = f(x_1 \sim x_n) \Rightarrow \frac{\partial y_1}{\partial x_1}, \frac{\partial y_1}{\partial x_2}, \cdots, \frac{\partial y_1}{\partial x_n}$

$\Rightarrow y_2 = f(x_1 \sim x_n) \Rightarrow \frac{\partial y_2}{\partial x_1}, \frac{\partial y_2}{\partial x_2}, \cdots, \frac{\partial y_2}{\partial x_n}$

$\frac{\partial y_m}{\partial x_1}, \frac{\partial y_m}{\partial x_2}, \cdots, \frac{\partial y_m}{\partial x_n}$

$\begin{bmatrix} a_{11}, a_{12}, \cdots, a_{1n} \\ a_{21}, a_{22}, \cdots, a_{2n} \\ \vdots \\ a_{m1}, a_{m2}, \cdots, a_{mn} \end{bmatrix} = A$

Jacobian

$y = Ax$

② $\alpha = y^T A \cdot x, \; y \in R^m, \; A \in R^{m \times n}, \; x \in R^n$

i) $\frac{\partial \alpha}{\partial x} = y^T \cdot A$

ii) $\frac{\partial \alpha}{\partial y} = \frac{\partial y^T A x}{\partial y} = \frac{\partial y^T (Ax)}{\partial y} = \frac{\partial (Ax)^T y}{\partial y} = x^T \cdot A^T$

$\hookrightarrow \frac{\partial y^T A x}{\partial y^T} = (Ax)^T = x^T \cdot A^T$

③ $\alpha = x^T A x, \quad A \in R^{n \times n}$ (square)

$\frac{\partial \alpha}{\partial x} = \frac{\partial (x^T A x)}{\partial x} = \frac{\partial (x^T A x)}{\partial x^T} + \frac{\partial (x^T A x)}{\partial x}$

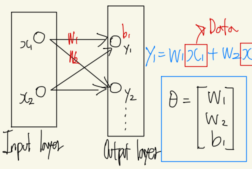
$= (Ax)^T + x^T \cdot A$

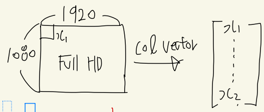
$= x^T \cdot A^T + x^T \cdot A = x^T (A + A^T)$

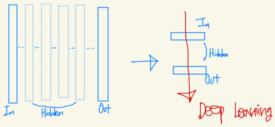
<인공지능 4주차 이론>

Perception (인간의 뉴런을 묘사) 다중입력 → 단일출력

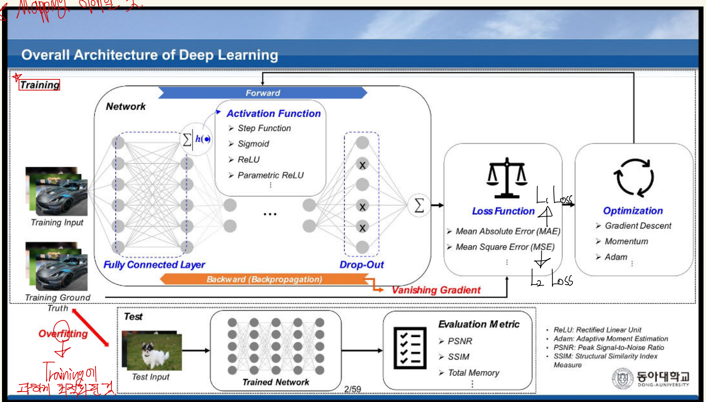
각각의 노드는 고유의 바이어스를 가짐







★ Mapping 이해할 것.



Learning

Supervised learning (지도학습)

⇒ 선생님이 있다 ⇒ 답이 있다

↳ Label, Ground Truth

Training Data ⟷(분리) Test Data

ex) 95% 70%

85% 90% ✓ Test 입장에서 정확도가 높으면 좋음

<인공지능 학습방법>

기호주의 인공지능 (컴퓨터 같이 기호 관계를 정의해 학습해야한다)

연결주의 인공지능 (인공 신경망으로 학습해야한다)

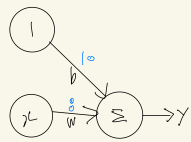
ex) Pixel

고양이의 RGB 값과 주변 배경의 RGB 값이 같으면 고양이로 구분 못하는 오류가 생김.

↓

XOR문제 제기

추론 : Inference



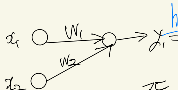
(perceptron 구조)

Supervised learning

→ Regression (Output : real Number) — 값을 추정

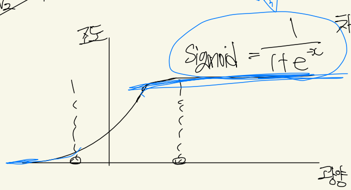
→ Classification
- Binary Classification (Output : 0 or 1)
- Muti-label 〃 (Output : 0, 1, 2, …, N)

h → Activation Fuction



$y_1 = h(W_1 x_1 + W_2 x_2 + b_1)$

ex) 빛이 강한 곳에서 빛을 조금 줄이거나 키운다고 차이를 구별할 수 있는가? → NO



핵심

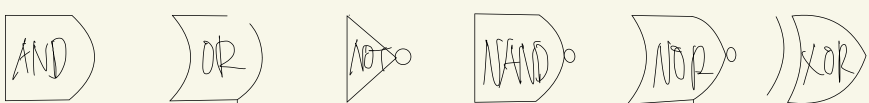
우리가 구하고자 하는 것은 어떤 Network가 있을 때 perceptron을 통해서 W값과 bias값을 구해야한다.

어떻게? Loss를 최소화할 수 있는 방법으로

어떤 방법으로? Backpropagation으로.

Overfitting을 유의하면서.

Logic gate



AND

0 0	0
0 1	0
1 0	0
1 1	1

OR

0 0	0
0 1	1
1 0	1
1 1	1

NOT

0	1
1	0

NAND

0 0	1
0 1	1
1 0	1
1 1	0

NOR

0 0	1
0 1	0
1 0	0
1 1	0

XOR

0 0	0
0 1	1
1 0	1
1 1	0

$W^T \cdot x$

$(m \times n)$ $(m \times 1)$

$(n \times 1)$

$\mathbb{R}$

$x \in \mathbb{R}^m$ $g \in \mathbb{R}^n$ $b \in \mathbb{R}^n$

$W \in \mathbb{R}^{m \times n}$

$W^T \in \mathbb{R}^{n \times m}$

$Loss \sum_{i=1}^{n} g^2$

$\frac{\partial L}{\partial g} = 2g \in \mathbb{R}^n$

<인공지능 5주차 이론>

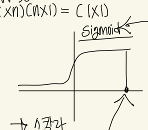
MLP (Multi Layer Perceptron)

$y = W^T x$

$x \in \mathbb{R}^n$ $\quad W \in \mathbb{R}^n$ → Input Node의 개수만큼 Weight 값은 따라 붙어야 함.

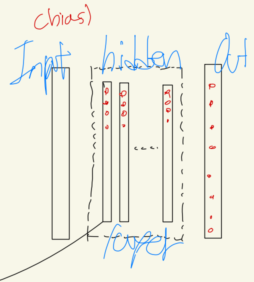
$= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} \quad = \begin{bmatrix} W_{11} \\ \vdots \\ W_{n1} \end{bmatrix}$

$W^T x$
$(1\times n)(n\times 1) = (1\times 1)$



$y = W^T x + b$ → 스칼라

$y = h(W^T x + b)$



MLP
= fully connected layer
= FC layer

N·P : W·b
H·P : 사람이 결정
opt.
Best-effort 방식 (노가다)

ex) Hidden layer 3개를 가지는.

$x \in \mathbb{R}^m \quad g \in \mathbb{R}^n \quad p \in \mathbb{R}^s$

$x \longrightarrow g \longrightarrow p$

$x_1 \xrightarrow{W_{11}} g_1 \qquad p_1$
$x_2 \quad W_{12} \quad g_2 \qquad p_2$
$\vdots \quad W_{1n} \quad \vdots \qquad \vdots$
$x_m \xrightarrow{W_{mn}} g_n \qquad p_s$

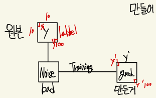
$W_1 \in \mathbb{R}^{m\times n}$, $b_1 \in \mathbb{R}^n$

$W_2 \in \mathbb{R}^{n\times s}$, $b_2 \in \mathbb{R}^s$

i) $g = h[\underset{(n\times m)}{W_1^T}\underset{(m\times 1)}{x} + \underset{(n\times 1)}{b_1}] \in \mathbb{R}^n$

ii) $p = h[\underset{(s\times n)}{W_2^T}\underset{(n\times 1)}{g} + \underset{(s\times 1)}{b_2}] \in \mathbb{R}^s$

iii) $p = h[W_2^T[h(W_1^T x + b_1)] + b_2]$



만들어 낸 것과 원본의 차이가 작으면 작을수록 좋음

$Loss = \sum_{i=1}^{100} |y_i - y'_i| = L_1\ Loss$
$= \sum (y_i - y'_i)^2 = L_2\ Loss$

무엇을 쓸지 정해져있지 않음
다 해봐야 함. Best-effort

경우에 따라 L_1, L_2 사용이 다르지만
Training이 잘된다.
↳ 정확도↑
↳ 속도가 빠르면 좋음
↳ w, b 같은 Parameter 값이 작으면 좋음
↳ 너무 작으면 성능↓ 즉, 잘조율해야함.

L_1 Loss = MAE (평균 절대 오차)
L_2 Loss = MSE (평균 제곱 오차)

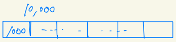
CEE (엔트로피)
↳ 불확실성을 수치화
↳ y · y' CEE
원본 y와 만들어낸 y'의 Entropy
→ y'이 y에 가까이 가게 곱하는 Entropy
↳ 작으면 작을수록 불확실성↓
↳ y'이 y에 가까이 감.

활성화 함수 (Activation function)

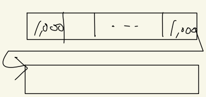
함수 로 이어지

Network parameter = 2.

- Forward Propagation
- Backward Propagation



Batch
- full batch (10,000)
- Mini batch ⇒ ex) 1,000 → Iteration → 1 epoch



10번 Iteration → 1 epoch

" → 2 epoch

Full batch (10,000)를 Mini batch (1,000)로 나눠서 Iteration 하면 10번이고

10번의 Iteration이 끝나면 1 epoch 이라고 한다.

이때 Mini batch, epoch은 Hyper parameter 즉, 사용자가 직접 정함.